Lecture 22

Theta functions and Poisson summation

Objectives. *Summing a rapidly decreasing function over integer translates produces a periodic function. Its Fourier coefficients prove Poisson summation, the result deferred from Lecture 7. Applied to a Gaussian, it gives the theta modular law. We also prove quasi-periodicity, count the zeros and derive the heat equation in the same normalisation.*

By the end of this lecture you will be able to:

- Prove Poisson summation for Schwartz functions with the course constants.
- Derive theta quasi-periodicity and identify all its simple zeros.
- Verify the heat equation by justified termwise differentiation.
- Derive the full modular law and specify its square-root branch.

22.1 Poisson summation

Recall that a Schwartz function is smooth and every derivative decreases faster than every reciprocal power. For $f \in \mathcal{S}(\mathbb{R})$ its *periodisation* is

$$p_f(x) = \sum_{n\in\mathbb{Z}} f(x+n), \qquad x \in \mathbb{R}.$$

For every derivative order j and every $N > 1$, Schwartz decay bounds $|f^{(j)}(x+n)|$ on $0 \le x \le 1$ by $C_{j,N}(1+|n|)^{-N}$. The M-test gives uniform convergence of the series and of each derivative series there. Thus p_f is smooth and has period one by reindexing.

Theorem 22.1 (Poisson summation). *For $f \in \mathcal{S}(\mathbb{R})$,*

$$\sum_{n\in\mathbb{Z}} f(x+n) = \sqrt{2\pi} \sum_{m\in\mathbb{Z}} \widehat{f}(2\pi m) e^{2\pi i m x}, \qquad x \in \mathbb{R}. \tag{22.1}$$

Both series converge absolutely and uniformly on $[0,1]$. In particular $\sum_n f(n) = \sqrt{2\pi}\sum_m \widehat{f}(2\pi m)$.

Proof. Define the ordinary period-one Fourier coefficient by

$$a_m = \int_0^1 p_f(x) e^{-2\pi i m x}\,\mathrm{d}x.$$

The integrand has period one, so this coefficient can be computed over any interval of length one. Absolute summability of the integrals follows from $\sum_n \int_0^1 |f(x+n)|\,\mathrm{d}x = \|f\|_1$.

Termwise integration and $u = x + n$ therefore give

$$a_m = \sum_n \int_n^{n+1} f(u) e^{-2\pi i m u}\,\mathrm{d}u = \int_{\mathbb{R}} f(u) e^{-2\pi i m u}\,\mathrm{d}u = \sqrt{2\pi}\,\widehat{f}(2\pi m).$$

The last equality uses the transform of Lecture 7, with kernel $(2\pi)^{-1/2} e^{-\mathrm{i}ku}$. By Theorem 7.3, these coefficients decrease faster than every reciprocal power of $|m|$. Hence $q(x) = \sum_m a_m e^{2\pi i m x}$ is an absolutely uniformly convergent series. Uniform termwise integration and exponential orthogonality show that its coefficients are a_m.

Fourier uniqueness identifies q with p_f. To check conventions, the orthonormal coefficients c_m in Lecture 6 for the angle function $p_f(v/(2\pi))$ satisfy $c_m = \sqrt{2\pi} a_m$. The continuous difference $p_f - q$, rescaled in this way, has every coefficient zero. Its Fejér sums are zero and converge uniformly to the difference by Theorem 6.2. Thus the difference vanishes. This proves the formula without using the sketched Dirichlet theorem. ∎

Stein–Shakarchi I, pp. 153–155, uses this periodisation proof. [1] For a positive period L, apply the theorem to $u \mapsto f(x + Lu)$. A change of variable in its Fourier integral gives

$$\sum_n f(x + nL) = \frac{\sqrt{2\pi}}{L} \sum_m \widehat{f}(2\pi m/L) e^{2\pi i m x/L}. \tag{22.2}$$

The factor $1/L$ is the Jacobian of that change of variable.

22.2 The theta series

For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$, define

$$\theta(z \mid \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}. \tag{22.3}$$

This is the *genus-one theta function*. Both arguments are always shown. It is distinct from the pendulum angle of Lecture 21 and the Heaviside function of Lecture 8. If associated with Lecture 20's lattice, its parameter is $\tau = \omega_3/\omega_1$. The normalised lattice generators are 1 and τ, corresponding to division of the original lattice coordinate by $2\omega_1$.

Proposition 22.2. *The function is entire in z and holomorphic in $\tau \in \mathbb{H}$. It is even in z, and it satisfies*

$$\theta(z + 1 \mid \tau) = \theta(z \mid \tau), \qquad \theta(z + \tau \mid \tau) = e^{-\pi i \tau - 2\pi i z} \theta(z \mid \tau). \tag{22.4}$$

The second relation is called quasi-periodicity.

Proof. On compact parameter sets, choose $\varepsilon > 0$ and $M > 0$ with $\operatorname{Im}\tau \geq \varepsilon$ and $|\operatorname{Im} z| \leq M$. The absolute value of a summand is at most $e^{-\pi\varepsilon n^2 + 2\pi M|n|}$, a summable sequence. Every mixed derivative only multiplies this by a polynomial in n, which is still summable. The M-test and the holomorphic-limit result in § 10.6 justify the holomorphy assertions and all these derivatives term by term.

Replace n by $-n$ for evenness. Period one follows from $e^{2\pi i n} = 1$. For the other relation, use

$$\pi i n^2 \tau + 2\pi i n(z + \tau) = \pi i (n+1)^2 \tau + 2\pi i (n+1) z - \pi i \tau - 2\pi i z.$$

The absolutely convergent sum can be reindexed by $n + 1$, giving (22.4). ∎

[1] Their transform is $\widehat{f}_{\mathrm{SS}}(\xi) = \int f(u) e^{-2\pi i \xi u}\,\mathrm{d}u$. It equals $\sqrt{2\pi}\,\widehat{f}(2\pi\xi)$ in our convention.

Unlike an elliptic function, this theta function is not periodic in its second lattice direction. The multiplier depends on z. The source normalisation is Stein–Shakarchi II, pp. 284–285. [2]

22.3 One simple zero per cell

Fix $\tau \in \mathbb{H}$. The theta function is not identically zero, because uniform integration of its series gives $\int_0^1 \theta(x \mid \tau)\,\mathrm{d}x = 1$. Its zeros are therefore isolated.

Theorem 22.3. *The zeros are exactly*

$$z = \frac{1+\tau}{2} + n + m\tau, \qquad n, m \in \mathbb{Z}.$$

Every zero is simple.

Proof. At $z_0 = (1+\tau)/2$ the series is $\sum_n (-1)^n e^{\pi \mathrm{i}(n^2+n)\tau}$. Pair n with $-n-1$. Their quadratic exponents agree and their signs are opposite, so the sum is zero. Quasi-periodicity shows the same for all the listed translates, since its multiplier never vanishes.

Where the denominator is nonzero, set $h(z) = \partial_z \theta(z \mid \tau)/\theta(z \mid \tau)$. Differentiating (22.4) and dividing gives

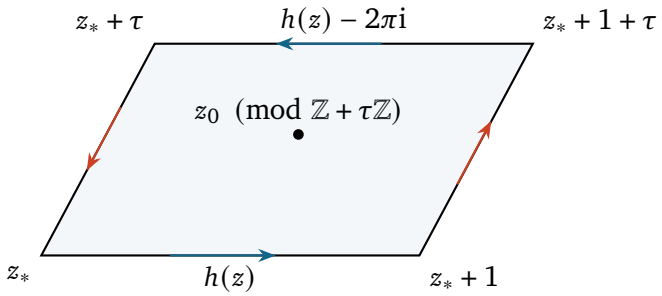
$$h(z+1) = h(z), \qquad h(z+\tau) = h(z) - 2\pi\mathrm{i}.$$

Choose a translated period cell with zero-free boundary. This is possible because an enlarged compact cell contains only finitely many zeros; a small generic translation moves all four edges off them. The two sides in the τ direction cancel in the integral of h. The bottom edge minus its translate at the top gives

$$\oint_{\partial P} h(z)\,\mathrm{d}z = \int_{z_*}^{z_*+1} \bigl(h(z) - h(z+\tau)\bigr)\,\mathrm{d}z = 2\pi\mathrm{i}.$$

The argument principle, Theorem 11.5, counts exactly one zero inside, with multiplicity. Every zero class modulo $\mathbb{Z}+\tau\mathbb{Z}$ has a representative in this cell, since the boundary contains no zero. The known class of z_0 must be that one class, and its multiplicity is one. This proves both completeness and simplicity of the list. ∎

The diagram displays the edge cancellation just used.



The boundary is positive and avoids the zeros. The slanted sides cancel; the quasi-periodic change along the horizontal pair contributes $2\pi\mathrm{i}$. The dot represents the unique simple zero in this shifted cell.

[2] Their capital $\Theta(z \mid \tau)$ is our $\theta(z \mid \tau)$. DLMF 20.2.3 uses $\theta_3(\pi z \mid \tau)$ for the same function, with nome $q = e^{\pi \mathrm{i}\tau}$. No other Jacobi theta family is used here.

22.4 The heat equation

The derivative convergence proved above permits differentiating every summand twice in z or once in τ. Their multipliers are $-4\pi^2 n^2$ and $\pi i n^2$, respectively. Hence

$$\partial_\tau \theta(z \mid \tau) = \frac{1}{4\pi i}\partial_z^2 \theta(z \mid \tau). \tag{22.5}$$

For real x and $t > 0$, define $u(x,t) = \theta(x \mid 4\pi i t)$. The chain rule gives $u_t = u_{xx}$ and the Fourier series is

$$u(x,t) = \sum_n e^{-4\pi^2 n^2 t} e^{2\pi i n x}.$$

It is a solution of the heat equation with period one and integral one over a period.

For a concrete second description, let $f_t(y) = (4\pi t)^{-1/2} e^{-y^2/(4t)}$. The real Gaussian transform, Theorem 4.5, gives $\widehat{f_t}(k) = (2\pi)^{-1/2} e^{-tk^2}$ after scaling. Poisson summation therefore identifies

$$u(x,t) = \frac{1}{\sqrt{4\pi t}} \sum_n e^{-(x+n)^2/(4t)}.$$

Thus the periodic heat solution is also a sum of translated real Gaussian kernels. Both descriptions follow from convergent series; no initial Dirac distribution is needed for this calculation.

22.5 The modular law

The transformation $\tau \mapsto -1/\tau$ maps $\mathbb{H}$ to itself and exchanges the two lattice directions after rescaling. Its theta identity is the *modular law* in our normalisation.

Theorem 22.4. *For $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$,*

$$\theta\left(\frac{z}{\tau} \,\middle|\, -\frac{1}{\tau}\right) = \sqrt{\frac{\tau}{i}}\, e^{\pi i z^2/\tau} \theta(z \mid \tau). \tag{22.6}$$

The square root is principal. In particular it equals $\sqrt{t}$ when $\tau = it$ with $t > 0$.

Proof. First let $t > 0$ and $z \in \mathbb{R}$. The function $f(u) = e^{-\pi t u^2 + 2\pi i z u}$ is Schwartz. The real Gaussian transform and modulation, followed by evaluation at $2\pi m$, give

$$\widehat{f}(k) = \frac{1}{\sqrt{2\pi t}} e^{-(k-2\pi z)^2/(4\pi t)},$$
$$\sqrt{2\pi}\,\widehat{f}(2\pi m) = \frac{1}{\sqrt{t}} e^{-\pi (m-z)^2/t}.$$

For the scaling, substitute $v = \sqrt{2\pi t}\, u$ in the Gaussian Fourier integral of Theorem 4.5; modulation replaces k by $k - 2\pi z$. Poisson summation at zero now yields

$$\theta(z \mid it) = \frac{1}{\sqrt{t}} \sum_m e^{-\pi (m-z)^2/t} = \frac{e^{-\pi z^2/t}}{\sqrt{t}} \theta\left(\frac{z}{it} \,\middle|\, \frac{i}{t}\right).$$

The last equality expands $(m-z)^2$ and uses $2\pi i m z/(it) = 2\pi m z/t$. Rearranging proves (22.6) for $\tau = it$ and real z.

For each such t, both sides are entire functions of z. The identity theorem, Theorem 10.6, extends the identity from real z to every complex z. Now fix any complex z. Both sides are holomorphic in $\tau \in \mathbb{H}$. For the composed theta series this follows from the same compact-set estimate as before: $-1/\tau$ stays in a compact subset of $\mathbb{H}$ and z/τ stays bounded on a compact τ set. Also τ/i lies in the right half-plane, so its principal square root is holomorphic there and positive at $\tau = \mathrm{i}t$. The identity theorem in τ, applied to the positive imaginary axis, proves the formula throughout $\mathbb{H}$. ∎

At $z = 0$ this becomes

$$\theta(0 \mid -1/\tau) = \sqrt{\tau/\mathrm{i}}\,\theta(0 \mid \tau).$$

For $\tau = \mathrm{i}t$ this gives the Gaussian theta identity deferred from Lecture 7:

$$\sum_{n\in\mathbb{Z}} e^{-\pi n^2 t} = t^{-1/2} \sum_{m\in\mathbb{Z}} e^{-\pi m^2/t}, \qquad t > 0.$$

For example, $\tau = 4\mathrm{i}$ gives $\theta(0 \mid \mathrm{i}/4) = 2\theta(0 \mid 4\mathrm{i})$. This fixes the direction of the square-root factor without evaluating either series numerically. Stein–Shakarchi II, Theorem 1.6, p. 290, gives the same full law. [3]

Summary. Periodisation and Fourier uniqueness prove Poisson summation for every Schwartz function. The theta series then has one simple zero per period cell, satisfies the heat equation and obeys the full modular law.

- The Poisson factor is $\sqrt{2\pi}$ and the sampled wave numbers are $2\pi m$ in the course transform convention.
- Theta is quasi-periodic, with zeros at $(1+\tau)/2 + \mathbb{Z} + \tau\mathbb{Z}$.
- The root in the modular law is principal and positive for $\tau/\mathrm{i} > 0$.

References

Stein–Shakarchi I, *Fourier Analysis*, pp. 153–156, and II, *Complex Analysis*, pp. 284–291. DLMF 20.2.3, 20.7.32 and 20.13.1 use the argument πz relative to ours; their heat coefficient changes accordingly. No triple-product theorem or higher-genus theta formula is imported here.

[3] Their printed law has left side $\Theta(z \mid -1/\tau)$ and right side $\sqrt{\tau/\mathrm{i}}\,\mathrm{i}e^{\pi \mathrm{i}\tau z^2}\Theta(z\tau \mid \tau)$. Substituting z/τ for their z gives ours. DLMF 20.7.32 uses argument πz and the principal root $(-\mathrm{i}\tau)^{1/2} = \sqrt{\tau/\mathrm{i}}$; evenness removes its transformed argument's minus sign.